

# VƯƠNG VIỆT HÙNG

## PERSONAL DETAILS

|            |                                                                            |              |                                                                                                                                                            |
|------------|----------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full name: | Vương Việt Hùng                                                            | Nationality: | Vietnamese                                                                                                                                                 |
| Phone:     | +84-583-885-391                                                            | Languages:   | Vietnamese (Native)                                                                                                                                        |
| Email:     | <a href="mailto:vuongviethung156@gmail.com">vuongviethung156@gmail.com</a> |              | English (IELTS 7.5)                                                                                                                                        |
| Residence: | Hanoi, Vietnam                                                             | ORCID:       |  <a href="https://orcid.org/0009-0006-8041-7608">0009-0006-8041-7608</a> |

## EDUCATIONAL QUALIFICATIONS AND ACADEMIC AWARDS

|                                                                                                                                                                                                                |                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Soongsil University</b>                                                                                                                                                                                     | Seoul, South Korea      |
| Master of Science in Engineering                                                                                                                                                                               | March 2024 – March 2026 |
| – School of Electronic Engineering                                                                                                                                                                             |                         |
| – GPA: 4.24/4.50                                                                                                                                                                                               |                         |
| – Thesis title: Online Continual Learning for LiDAR Panoptic Tracking                                                                                                                                          |                         |
| – Thesis length: 10,387 words                                                                                                                                                                                  |                         |
| – Thesis summary: Developed an online continual learning framework for LiDAR panoptic tracking, using scene flow-guided temporal association and self-training to adapt to sequentially changing environments. |                         |

|                                                                                                                                                                                                                     |                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Hanoi University of Science and Technology</b>                                                                                                                                                                   | Hanoi, Vietnam               |
| Bachelor's Degree in Control Engineering and Automation                                                                                                                                                             | August 2019 – September 2023 |
| – School of Electronic and Electrical Engineering                                                                                                                                                                   |                              |
| – GPA: 3.34/4.00                                                                                                                                                                                                    |                              |
| – Thesis title: Application of Stereo Matching in SLAM for Autonomous Vehicles                                                                                                                                      |                              |
| – Thesis length: 13,536 words                                                                                                                                                                                       |                              |
| – Thesis summary: Investigated stereo matching for depth estimation and 3D reconstruction in autonomous-vehicle SLAM, implemented and evaluated classical and deep-learning-based dense stereo matching approaches. |                              |

## PROFESSIONAL AFFILIATIONS AND MEMBERSHIPS

None to date

## EMPLOYMENT HISTORY

|                                                                                                                                                                     |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>VinDynamics</b>                                                                                                                                                  | Hanoi, Vietnam          |
| SLAM & Navigation Engineer                                                                                                                                          | March 2026 – current    |
| – Worked on visual-inertial odometry and multi-sensor height mapping for real-time perception on humanoid robotic platforms.                                        |                         |
| – Contributed to VIO pipeline profiling and latency optimization, and extended GPU-based height mapping with LiDAR, depth camera, IMU, and joint-state integration. |                         |
| <b>ANDA Lab, Soongsil University</b>                                                                                                                                | Seoul, South Korea      |
| Research Assistant                                                                                                                                                  | March 2024 – March 2026 |
| – Conducted research on LiDAR panoptic segmentation and tracking, with a focus on domain adaptation and online continual learning.                                  |                         |

- Contributed to model development and evaluation, research publications, researcher mentoring, and technical seminars.

## Rikai Mind

AI Engineer Intern

Hanoi, Vietnam

August 2022 – January 2023

- Developed a computer vision system for object volume estimation from monocular images and camera–object distance, using CNN-based features and keypoint regression.
- Contributed to data collection, model development and evaluation, and deployment of the inference pipeline through a FastAPI-based service.

## OTHER RESEARCH EXPERIENCE

---

### Mandevices Laboratory, Hanoi University of Science and Technology

Research Assistant (Voluntary)

Hanoi, Vietnam

August 2020 – September 2023

- Conducted research on deep-learning-based bearing fault diagnosis, including simulation-based domain adaptation to address limited real-world training data.
- Contributed to data collection, model development and evaluation, research publications, and researcher mentoring.

## RESEARCH OUTPUTS

---

### Refereed journal articles

1. **Vuong, H. V., & Yoo, M.** (2026). Domain adaptive LiDAR panoptic segmentation via pseudo-supervised instance learning. *IEEE Transactions on Intelligent Transportation Systems*.  
<https://doi.org/10.1109/TITS.2026.3695516>  
Journal Impact Factor: 9.1 (2025, JCR)
  - Contribution: Designed and implemented the proposed methodology; designed and conducted the experiments; analyzed and interpreted the results; prepared the manuscript.
  - Research outcome: Proposed PILLA for instance-level pseudo-supervised domain adaptation, achieving 35.97% PQ on nuScenes and outperforming AdaptLPS by 13.14 percentage points.
2. Ha-Phan, N.-Q., **Vuong, H. V., & Yoo, M.** (2025). Instance embedding as queries for DETR-based LiDAR panoptic segmentation. *IEEE Transactions on Intelligent Vehicles*, 10(10), 4629–4641.  
<https://doi.org/10.1109/TIV.2024.3488035>  
Co-first authors: Ha-Phan, N.-Q. and Vuong, H. V.  
Journal Impact Factor: 14.3 (2025, JCR)
  - Contribution: Primarily designed and implemented the proposed methodology; conducted the experiments, analyzed and interpreted the results; prepared the manuscript with the co-first author.
  - Research outcome: Proposed a DETR-based architecture using instance embeddings as object queries, achieving 63.40% PQ on SemanticKITTI and 77.08% PQ on nuScenes, outperforming previous methods by 1.20 and 1.94 percentage points, respectively.
3. Nguyen, K. N. M., **Vuong, H. V.,** Ha-Phan, N.-Q., & Yoo, M. (2025). FuPaSCo: Long-range and local context fusion for 3D panoptic scene completion. *Image and Vision Computing*, 163, 105776.  
<https://doi.org/10.1016/j.imavis.2025.105776>  
Journal Impact Factor: 5.0 (2025, JCR)

- Contribution: Provided consultation on the methodology; assisted with the experiments by developing an evaluation tool for performance analysis; reviewed the manuscript and provided feedback during its preparation.
  - Research outcome: Developed FuPaSCo, a dual-branch CNN–Transformer framework for LiDAR panoptic scene completion, achieving 26.10% PQ<sup>†</sup> on SemanticKITTI and outperforming PaSCo by 2.49 percentage points.
4. Nguyen, T. H., **Vuong, V. H.**, Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). Generalized simulation-based domain adaptation approach for intelligent bearing fault diagnosis. *Arabian Journal for Science and Engineering*, 49(12), 16941–16957. <https://doi.org/10.1007/s13369-024-09282-1>  
Journal Impact Factor: 3.1 (2025, JCR)
- Contribution: Assisted with data collection; developed the bearing vibration signal model; conducted experiments combining the signal model with the baseline deep-learning approach; contributed to the analysis and interpretation of the experimental results.
  - Research outcome: Developed a simulation-based domain adaptation approach using synthetic bearing vibration signals, achieving 74–98% fault classification accuracy across different bearing types.

#### Published conference proceedings

1. Nguyen, T. H., Le, H. A., **Vuong, V. H.**, Tran, L. D. A., Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). ConvMamba: A data-efficient neural network for bearing fault diagnosis. In *2024 7th International Conference on Green Technology and Sustainable Development (GTSD)* (pp. 165–172). IEEE. <https://doi.org/10.1109/GTSD62346.2024.10674880>
- Contribution: Designed and implemented the proposed methodology; designed and conducted the experiments; analyzed and interpreted the results; prepared the manuscript with assistance from the co-authors.
  - Research outcome: Developed ConvMamba, a data-efficient architecture combining convolutional networks and Vision Mamba, improving classification accuracy by 27 percentage points over Hybrid-CNN-MLP on the CWRU bearing fault dataset.

#### Manuscripts under review

1. Pham, T. P., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). SM3-PSC: Sparse multi-modal Mamba for 3D panoptic scene completion with multi-scale feature refinement. *Manuscript submitted to IEEE Transactions on Image Processing*.
- Contribution: Provided consultation on the proposed methodology and reviewed the manuscript, provided technical feedback and suggestions during its preparation.
  - Research outcome: Developed SM3-PSC, a sparse multi-modal framework combining RGB and LiDAR features with Mamba-based context modeling, achieving 28.20% PQ<sup>†</sup> on SemanticKITTI while running the 3D pipeline 2.8× faster than FuPaSCo.
2. Nguyen, K. N. M., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). LW-PSC: Lightweight design for LiDAR panoptic scene completion. *Manuscript submitted to Applied Soft Computing*.
- Contribution: Provided consultation on the proposed methodology and reviewed the manuscript, provided technical feedback and suggestions during its preparation.
  - Research outcome: Developed LW-PSC, a lightweight BEV-based framework for LiDAR panoptic scene completion, achieving 13.59% PQ at 11.3 FPS while reducing the model size from 40.65M to 29.16M parameters.

## **RESEARCH GRANTS AND RELEVANT AWARDS**

---

None to date